# PFAS Estimator User Guide

Version 1.0

## Website
## https://pfasestimator.org

PFAS Estimator is a free, publicly accessible web application that translates EPA Fifth Unregulated Contaminant Monitoring Rule (UCMR 5) drinking water monitoring data into ZIP code-level environmental health information. The platform is intended for researchers, clinicians, public health professionals, and community members seeking evidence-based context for PFAS detections in community water systems.

## Key Features

• Search any U.S. ZIP code served by a monitored community water system.

• View detected PFAS compounds and co-occurring analytes.

• Compare observed concentrations with EPA Maximum Contaminant Levels (MCLs) or evidence-based reference levels.

• Read compound-specific summaries describing toxicological mechanisms, health effects, regulatory context, and clinical considerations.

• Access interactive maps, downloadable tables, and methodology documentation.

## Underlying Dataset

EPA UCMR 5 (2023–2025) monitoring data processed into a nationwide analytical dataset containing 40,311 compound-level records spanning 12 analytes across 14,071 ZIP codes served by 4,885 public water systems. The framework retains the highest observed concentration per compound per ZIP code to characterize the upper bound of observed drinking-water concentrations.

## Clinical Translation Framework

Each analyte is paired with a manually curated evidence profile synthesizing EPA documentation, National Academies guidance, and peer-reviewed toxicological and epidemiological literature. AI-generated summaries are grounded in this curated knowledge base to provide accessible, compound-specific context while preserving scientific accuracy.

## Availability

Free to use. No registration required. Intended for environmental exposure assessment, public health communication, education, and clinical context—not as a diagnostic or medical device.

## Suggested Citation

PFAS Estimator. https://pfasestimator.org. Associated manuscript: "PFAS Contamination in U.S. Community Water Systems: A National ZIP Code-Level Analysis of EPA UCMR 5 Data with a Clinical Translation Framework" (manuscript in preparation).